⇑Sagar Panchal
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About:

Software Engineer with hands-on machine operation experience. My career has spanned the full spectrum of Automation Engineering, from hands-on electrical testing and debugging to developing and maintaining critical components within large-scale codebases that powers complex machineries. Currently seeking opportunities to build impactful technologies while continuing to grow as an engineer and problem solver.

Education:

MS, Computer engineering (3.41/4) – Dec-17
New York University NYU
BE, Instrumentation & Controls Engineering (7.04/10) – Jun-14
Gujarat Technological University

Skills:

Automation:
- PC based machine automation systems (C,C++,C#,Java)
- Real-time operating system (VxWorks)
- Computer Vision (Keyence, Cognex & previously OpenCV)
- Control logic implementation, execution & testing
- Equipment troubleshooting & root cause analysis
- PLC based industrial automation systems (Ladder, FBD), HMI, SCADA

Software Development:
- Desktop - Java, C#, Python (AI-driven)
- Web - HTML, CSS, JS, DB - MDB, SQL
- Git(SourceTree, Bash), Jira, Bitbucket
- Maintaining large-scale & legacy codebases

Professional Experience:

Software Engineer III - Automation, Universal Instruments, NY — *Sep 22 - present*

Core Responsibilities:
- Maintain and evolve high-precision, high-speed software modules across the Fuzion SMT platform, ensuring reliability, performance, and production readiness.
- Lead remote production-line diagnostics by analyzing logs, replicating site conditions, performing root-cause investigations and delivering targeted fixes, patches, and release updates.
- Conduct code reviews and enforce software-quality standards across the platform, ensuring robust design, maintainability and consistency in mission-critical automation workflows.
- Enable field and application engineers to successfully deploy and validate applications by providing technical guidance, interpreting machine data, and supplying required configurations or software updates.
- Replicate site issues and lead diagnosis to reach the root cause of problems.

Key contributions:
- Designed, developed, and delivered Selective Placement, Fuzion's first smart pick/buffer/place workflow using customer provided categories and avoiding-priorities, improving product quality and placement intelligence.
- Engineered SBS using Keyence SR-X300, delivering Fuzion's first side-barcode scanning capability, making fuzion scan barcode in all the three X, Y and Z axes, with live-view, traceability, and full error-recovery retuning support.
- Upgraded the AMS workflow from a single point to 13 wafer-level locations, improving calibration robustness and hence maintaining placement accuracy.
- Built the wafer-to-board mapping system that converts customer wafer maps into Fuzion board templates, eliminating hours of manual template creation and enabling direct wafer-data usage.
- Lead Initial implementation of AVP (Advanced Vision Processing), enabling user-defined fiducial finds across ECS, NPI, and Production workflows, increasing alignment flexibility and reducing setup constraints.
- Integrated Cognex QuickBuild (VPP) with MCOS↔QBIT TCP/IP triggers and full reject/retry handling, enabling Dispense Quality Inspection on the Precision-Bonder platform for the first time.
- Delivered multiple platform-level enhancements including the Light Intensity Controller, nested-tray support, isosceles fiducial and configurable park-position, strengthening core system capability.

Automation Engineer , Nor-Cal controls, Ca — *Jan - Jul 22*
- Designed SCADA interfaces and integrated PLC logic with inverter and sensor data to deliver reliable solar power-plant automation including FAT and commissioning support.
- Tech stack: Ignition SCADA · GE PLC · SEL-RTAC · Red Lion

Software Engineer - Machine Automation, Mini-Circuits, NY — *Feb 18 - Nov 21*
- Evolved machine-automation software across rotary machines with CV Inspection coordinating servo-motion, sensor-driven logic, EVT automation via PCI-e card and PC application based database-supported workflows.
- Enhanced precision laser-measurement stations, semiconductor-packaging machines, and PID-controlled TVAC systems by integrating database-linked classification logic, improving QA reliability and reducing manual effort.
- Helped support modernization of full-stack systems including the public shopping-cart redesign, product-visibility management and lifecycle-tracking tools, increasing platform reliability and removing hours of manual data handling.

Developer Intern, RevMax Fleet Optimization, NY based startup* — *Fall 2017*
- Developed Doppler-radar based pedestrian-counting device and transferred data to back-end services using BLE.
- Deployed sensors at multiple NYC locations and validated field performance

Instructor (ITEST Program), robotics & mechatronics dpt., NYU — *Fall 2017*
- Mentored high-school robotics teams in autonomous-robot workflows for FIRST-Robotics/VEX competitions.
- Taught foundational programming concepts in Arduino & sensor-integration to student robotics teams.

Lead Instructor (CrEST Program), Center for K12 STEM Education, NYU — *Summer 2017*
- Mentored high-school interns in Arduino, IoT, robotics, and 3D printing & led STEM projects including:
  -Autonomous Car (Mouse in a Maze) - STM32 Board: https://github.com/sap586/RTES_Project
  -Wireless Robot Controller - Arduino: https://github.com/sap586/WirelessRobotDay5Activity

Automation Engineer, Prima Automation, Ahmedabad, IN — *Jun 14 - Jan 16*
- Worked with machine-automation systems using Siemens PLC-HMI as control module, provide technical support including software development, testing, FAT/SAT and commisioning support.
- Support development and maintenance of software (like HMI screens, PLC code, AC-drive configurations, IP-configs, etc) and documentation including control panel layouts made in AutoCAD and electrical drawings made in E-plan.